

Leveraging Information Retrieval and Large Language Models for Personalised Legal Document Search in International Law

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1 Chapter One

1.1 Introduction

1.1.1 Problem Statement:

Retrieving information from legal documents is the challenge legal practitioners constantly face due to diversity of legal systems across different jurisdictions, complexity, and large volume of the documents. Access to accurate, relevant, customised/personalised, and comprehensive legal information is crucial and timesaving for legal practitioners (lawyers), scholars, and policy makers. Effective systems for legal information retrieval allows users to efficiently find relevant treaties, case laws, and legal decisions facilitating timely and informed decision-making and research.¹ Traditional Information Retrieval (IR) systems struggle to interpret jurisdiction-specific terminology or critically analyse that leads to irrelevant results. They do not tailor results according to individual needs or preferences, which impacts efficiency. Also, these systems provide non-legal information, which are unnecessary and inadequate for the expected search result.² Large Language Models (LLMs) are advanced Artificial Intelligence (AI) systems that could be trained on extensive text datasets to generate personalised contents tailored to individual needs, summarise, and analyse textual data.³

1.1.2 Objective:

This project aims to develop a personalised legal document retrieval system by leveraging the LLMs to create a platform that tailors search results to the specific individual needs and preferences, enhances efficiency, accuracy, and relevance. By leveraging LLMs, the system can incorporate contextual understanding, intuitive explanation, and advanced relevance ranking that can address challenges faced by traditional IR systems. The LLMs provide justification for each ranking and a general prediction about the subject matter legal documents. The focus would be on

¹ Weicong Qin, et al. (2024). SIGIR 2024: The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval. Washington DC USA, July 14 – 18. New York: Association for Computing Machinery. Available at: <https://doi.org/10.1145/3626772.3657717> (Accessed: 11 July 2024)

² Ganguly, D. et al. (2023). 'Legal IR and NLP: The History, Challenges, and State-of-the-Art', Advances in Information Retrieval. Dublin, Ireland, 2 – 6 April. Switzerland: Springer. https://doi.org/10.1007/978-3-031-28241-6_34

³ Mims, C. (2024). 'A Powerful AI Breakthrough Is About to Transform the World', The Wall Street Journal, 15 Nov. Available at: <https://www.wsj.com/tech/ai/a-powerful-ai-breakthrough-is-about-to-transform-the-world-095b81ea> (Accessed: 15 November 2024)

international law datasets accessible online on different international law platforms, specifically in this project, the World Trade Organisation (WTO) database.

1.1.3 Research Questions:

To achieve the objective, set in this project, it will strive to find answers to two research questions:

1. How can (international) legal systems leverage traditional IR systems to search legal documents?

This will explore the performance of IR systems to deliver relevance ranking and tailored results based on a query.

2. How effectively can LLMs generate explainable and personalised outputs to justify document relevance and predict outcomes?

This will explore how LLMs can provide natural language explanations in justifying the reason for highly ranked legal documents/information and provide a prediction for a similar situation.

Therefore, this thesis project is outlined in three chapters; (1) Chapter One: Introduction, Literature Review, Methodology, and Dataset; (2) Chapter Two: Baseline IR System; (3) Chapter Three: LLM.

This project will discuss the limitations, further work, and a conclusion as well.

A bibliography of references and appendices of applied codes will be provided in the end.

1.2 Literature Review

Legal systems vary depending different jurisdictions and subject matter aspects of law. Almost all legal systems (including local and international), jurisdictions, and law aspects preserve past provisions as the central and main source of functioning. They rely heavily on past provisions to manage current and future legal decisions. Past provisions lay the foundation for predictable, consistent, and evolving legal decisions. They ensure certainty, fairness, and stability.⁴ International and local legal systems store past provisions in different formats as the basis of legal precedents. Case laws form a common format of stored past provisions. These case laws are extremely crucial for legal practitioners such as lawyers, policy makers or law scholars. A lawyer needs case laws to provide sound legal advice, argue cases effectively, and interpret laws based on their past applications. Similarly, scholars and policy makers need case laws to critically analyse their effectiveness in the past to guide legal reforms and ensure legal continuity in the future.⁵ Among millions of case laws in different systems and jurisdictions, selectivity and research ability of legal practitioners to retrieve information is vital. A customised/personalised information retrieval could lead to more informed, timely, and accurate decisions by legal practitioners. It makes legal research more efficient, relevant, and tailored to the needs of the user.⁶

In this cutting-edge era of AI, legal practice could massively leverage traditional IR systems and modern neural network-based algorithms to access comprehensive, efficient, personalised and/or tailored information in seconds.

Legal systems have relied on traditional IR techniques such as inverted indexes, lexical ranking algorithms (e.g. BM25), etc in the past. Many modern legal databases still utilise these IR systems to leverage keyword-based search with complex query operators that could provide the user with efficient, transparent, scalable results. They can swiftly rank millions of legal documents – an analytical strength matching the fast-

⁴ Howe, M.L. & Knott, L. M. (2015) 'The Fallibility of Memory in Judicial Processes: Lessons from the Past and Their Modern Consequences', *Behavioural Sciences*, 23(5), pp. 633 – 656.
<https://doi.org/10.1080/09658211.2015.1010709>

⁵ Merryman, J. & Perez-Perdomo, R. (2018) *The Civil Law Tradition: An Introduction to the Legal Systems of Europe and Latin America*. Stanford University Press.

⁶ Kim, MY. Rabelo, J. Okeke, K. *et al.* (2022) 'Legal Information Retrieval and Entailment Based on BM25, Transformer and Semantic Thesaurus Methods', *Rev Socio-network Strat* 16, 157–174.
<https://doi.org/10.1007/s12626-022-00103-1>

growing volume and complex literature of legal documents.⁷ Meanwhile, there are challenges associated with these traditional IR systems. “Legal relevance is a nuanced concept: a legal document’s importance is not determined by topical similarity alone, but by legal context, authority and applicability”.⁸ In this sense, the traditional IR systems are focusing on keyword overlap rather than a true relevance in terms of legal meaning or interpretation. However, research indicates, in the past five years, the traditional IR systems are augmented rather than being abandoned. “This is probably because they are strong building blocks whose strength (speed, precision control, and simplicity) should be preserved”. Therefore, we will dive more into the logic of their performance in this piece of research.⁹

BM25 is a key-word matching method that has served as a base for IR systems from long ago. “It uses term frequencies and document length normalisation to score documents, excelling at precision when query terms exactly match document terms.” BM25 works simply and does not require any training data, however, it can only retrieve and rank the documents that exactly match the query’s keywords letting the documents with different vocabulary and relevant legal meaning behind. For example, if the query consists of a word such as “United States”, BM25 cannot capture documents only having “US” to retrieve and rank their data.¹⁰

Sentence-BERT produces fixed-size sentence embeddings that can capture semantic meaning. Sentence-BERT processes two sentences by the same BERT-based encoder called twin network and the embeddings results are compared through cosine similarity to determine semantic meaning. By this performance, Sentence-BERT overcomes the challenge of keyword overlap. For example, if the query uses the word “movie”, Sentence-BERT can capture the word “film” based on semantic meaning retrieval and ranking in its result. Another aspect is that Sentence-BERT can work well on large documents and can find the most similar pair in 10,000 sentences just in seconds by embedding approach that could produce reliable outcome. At the same

⁷ Wiggers, G. Verberne, S. (2023) ‘Bibliometric-enhanced Legal Information Retrieval: Combining Usage and Citations as Flavors of Impact Relevance’, *Journal of the Association for Information Science and Technology*, 74(8), pp. 1010-1025. <https://doi.org/10.1002/asi.24799>

⁸ Feng, Y. Li, C. Ng, V. (2024). ‘Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics’, Association for Computational Linguistics, 11 – 16 August.

⁹ Locke, D. Zuccon, G. (2022) ‘Case Law Retrieval: Problems, Methods, Challenges and Evaluations in the Last 20 Years’, *Computer Science*. <https://doi.org/10.48550/arXiv.2202.07209>

¹⁰ Saravanan, M. Ravindran, B. & Raman, S. (2009) ‘Improving Legal Information Retrieval Using an Ontological Framework’, *Artificial Intelligence and Law*, 17(2), pp. 101 – 124. <https://doi.org/10.1007/s10506-009-9075-y>

time, recent research shows that Sentence-BERT can sometimes underperforms BM25 on out-of-domain tasks due to sensitivity of semantic models to mismatched domains.¹¹

LLMs are extremely large neural network language models with millions to billions of parameters. The families include GPT-3, GPT-4, PaLM, and LLaMA. These models can demonstrate strong natural language understanding and generation capabilities in variety of tasks. They can display near human-like proficiency on given tasks like answering questions, summarisation, reasoning, prediction, etc. LLMs model complex dependencies in text by training on diverse data (books, articles, web information, etc). They not only learn word meanings but semantic relations, facts, and reasoning. The other impressive side is that they have the adaptation capability to new tasks with only a few examples in the prompt. It is worth mentioning that in retrieval and ranking tasks LLMs are mostly used as re-rankers due to being costly, having limited input size called the tokens. These tokens are probably a few thousands, depending on time scale (normally in a minute), which are called Token per Minute (TPM). However, they are the best in personalised analyses, predictions, and justifications.¹²

To conclude, each of the above-mentioned systems have their own specialities, which are worth using. Legal systems can leverage all of them either in singular and separate approaches or hybrid based on the results they demand. Legal systems can utilise their capability based on their specific characteristics and performance level. The methodology will discuss more the approach considered in this project to leverage traditional IR systems and LLMs for the benefit of international legal systems.

¹¹ Reimers, N. Gurevych, I. (2019) 'Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing', Association for Computational Linguistics, 3 – 7 November, pages 3982–3992, Hong Kong, China.

¹² Chang, Y. et al. (2024) 'A Survey on Evaluation of Large Language Models', *ACM Transactions on Intelligence Systems and Technology*, 15(3), pp. 1 – 45. <https://doi.org/10.1145/3641289>

1.3 Methodology

This project focuses on interdisciplinary research consisting of legal systems and applied AI elements. This research has used a small case law dataset from an international legal system to apply different AI methods and achieve the goal of information retrieval including ranking, justifying, and predicting. As already mentioned, three methods of information retrieval (BM25, Sentence-BERT, and LLMs) are applied on the WTO case law dataset. Before application of these three methods, a comprehensive data cleaning and preprocessing is also carried out on the dataset. Each of the methods evaluated based on their Precision, Recall, and Normalised Discounted Cumulative Gain (nDCG) performance. In terms of coding, different python-3 packages and libraries are used in this project, which will be discussed in detail in each section.

1.4 Dataset

1.4.1 Dataset Preparation:

The dataset used in this project is consisted of 111 summary case laws of WTO. These case laws are called “dispute cases” on WTO’s website. WTO is the only global organisation dealing with rules and disputes of trade between nations. Since 1995 to date, WTO has recorded 635 disputes from different countries around the world and has issued 350 rulings. To create the dataset for this project, only those dispute cases have been considered, which have a one-page summary of the original case law. The summary case laws have a similar format, each of them is consisted of:

- A short title including the name of the respondent country and the dispute topic.
- A unique reference number starting with “DS”.
- A table including the information about parties, the articles applied and agreed on, and the timeline of the dispute.
- Description of the measures/ rules applied.
- Summary of findings.
- Miscellaneous issues.

AUSTRALIA – SALMON (ARTICLE 21.5 – CANADA)¹
(DS18)

PARTIES		AGREEMENTS	TIMELINE OF THE DISPUTE	
Complainant	Canada	SPS Arts. 2.2, 2.3, 5.1, 5.5 and 5.6 DSU Art 10.3	Referred to the Original Panel	28 July 1999
			Circulation of Panel Report	18 February 2000
Respondent	Australia		Circulation of AB Report	NA
			Adoption	20 March 2000

Figure 1.1: Example of title and table of WTO case laws.¹³

The dataset created from summary case laws consists of 7 columns: Case ID, Title, Complainant, Respondent, Date, Articles, and Summary. The information in these columns is extracted from different parts of the summary case laws and saved in a “csv” file via python code by using packages such as “pdfplumber” and “pandas”. Libraries such as “re” and “os” are used to match the text pattern and handle file/directory too.

¹³ World Trade Organisation (1995) Documents and Resources. Available at: <https://www.wto.org/>

Case ID	Title	Complainant	Respondent	Date	Articles	Summary
Ds491	US – Coated Paper (Indonesia)	Indonesia	United States	22 January 2018	ADA Arts. 3.5, 3.7, 3.8. ASCM Arts. 2.1(c), 12.7, 14(d), 15.5, 15.7, 15.8	FINDINGS2 • ASCM Art. 14(d) (rejection of in-country prices as benchmarks): The Panel rejected Indonesia's argument that the USDOC improperly concluded that Indonesian prices for standing timber and logs were unusable as benchmarks....

Table 1.1: First row of the dataset.

1.4.2 Dataset Cleaning:

To clean the dataset before implementing the models, missing and duplicate data in each column has been searched via Python coding using “pandas”. For some of the columns such as complainant, respondent, date, and articles, it is normal to be duplicated because the name of countries involved, and articles used to resolve the dispute could be similar in many cases and coincidentally some dates might be the same as well. These duplicates are not checked in python coding, however, the accuracy of the dataset for such cases is checked manually. Meanwhile, the number of duplicates and missing data in other columns (Case ID, Title, and Summary) are zero. Therefore, to perform a proper data cleaning, a second extraction process applied to reduce the missing data, and the remaining blank data (only in Date column) is replaced by “Unknown”. After this performance, the missing value in the whole dataset is reduced to 0.

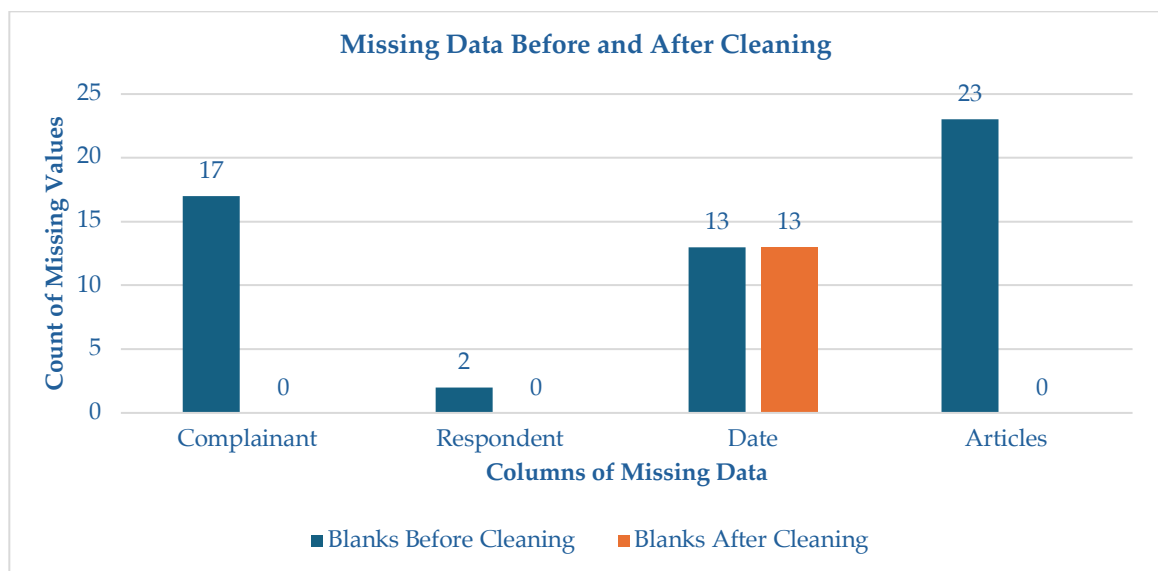


Chart 1.1: Visualisation of missing data before & after cleaning.

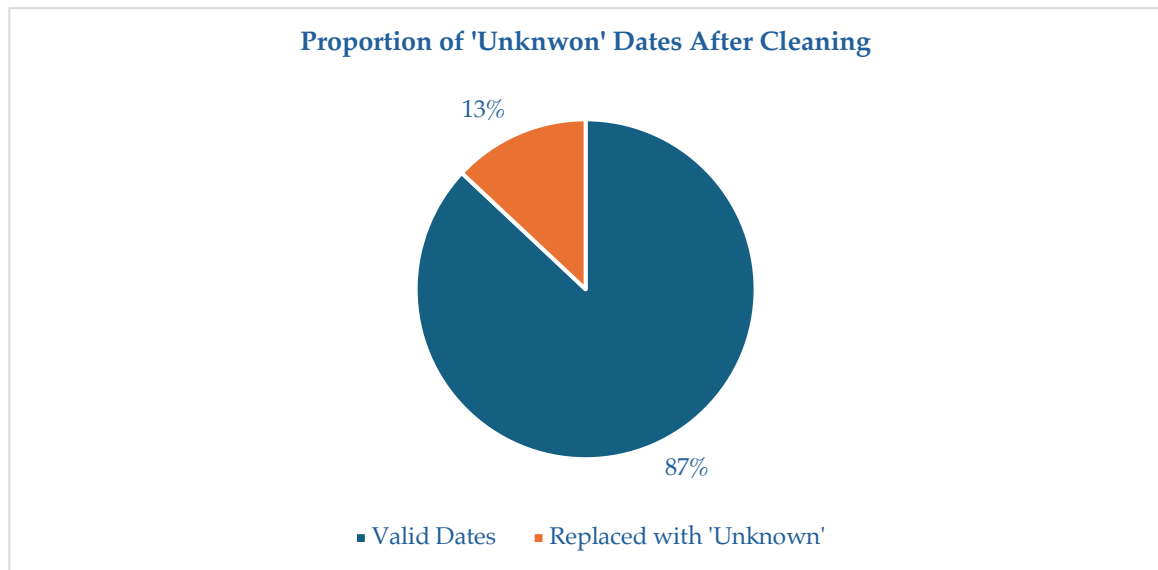


Chart 1.2: Proportion of "Unknown" dates after cleaning.

1.4.3 Dataset Preprocessing:

The dataset is pre-processed before implementing the models. Python packages like "nltk", "spacy", "contractions", and "sentence-transformers" are used for preprocessing purposes. The preprocessing includes normalisation, (such as expanding contractions, converting to lowercase, standardizing legal references, and removing punctuations & special characters), tokenisation (including stop-word removal), lemmatisation, sentence segmentation, and generation of embeddings. Before applying the mentioned preprocessing steps, a standard format of (YYYY-MM-DD) applied on the date column using "pandas" library. The normalisation is applied only on three columns (Title, Articles, and Summary) which required preprocessing. Packages and libraries such as "re", "contractions", and "pandas" used in normalising the dataset. Tokenisation also applied on normalised columns in order to remove stop-words and tokenise the text using libraries of "nltk" package. Below table visualises the changes in the columns after applying normalisation & tokenisation (only first 4 rows).

Title (Original)	Title (Tokenized)	Articles (Original)	Articles (Tokenized)	Summary (Original)	Summary (Tokenized)
US – COATED PAPER (INDONESIA)1	us coated paper indonesia1	ADA Arts. 3.5, 3.7, 3.8; ASCM Arts. 2.1(c), 1.2.7, 14(d), 15.5, 15(c)	ada arts 35 37 38 ascm arts 21c 127 14d 155 15c	FINDINGS2 • ASCM Art. 14(d) (rejection of in-country prices)	findings2 ascm art 14d rejection incountry prices
US – WOOL SHIRTS AND BLOUSES1	us wool shirts blouses1	ATC Arts. 6 and 2.4	atc arts 6 24	/ AB FINDINGS • ATC Art. 6 (transitional safeguard mechanism)	ab findings atc art 6 transitional safeguard mechanism
PHILIPPINES – DISTILLED SPIRITS1	philippines distilled spirits1	GATT Art. III:2, first and second sentences	gatt art iii2 first second sentences	/ AB FINDINGS • GATT Art. III:2 (national treatment, trade rules)	ab findings gatt art iii2 national treatment trade rules
US – UPLAND COTTON (ARTICLE 21.5 – BRAZIL)1	us upland cotton article 215 brazil1	ASCM Arts.3, 5(c), 6.3(c), item (j) of illustrative list; AA Arts. 10.1 and 8	ascm arts3 5c 63c item j illustrative list aa arts 101 8	/ AB FINDINGS • AA Arts. 10.1 and 8, and ASCM Arts. 3.1(a), 3.2; illustrative list	ab findings aa arts 101 8 ascm arts 31a 32 illustrative list

Table1.2: Visualisation of dataset columns before & after normalisation & tokenisation.

As seen in the table, punctuations, any special character, and stop words are removed. Text is converted to lowercase, contractions expanded, and the legal references are standardised. The bar chart below demonstrates how the text length reduced after tokenisation and normalisation. The bar chart below only considers first 5 rows of the dataset as an example.

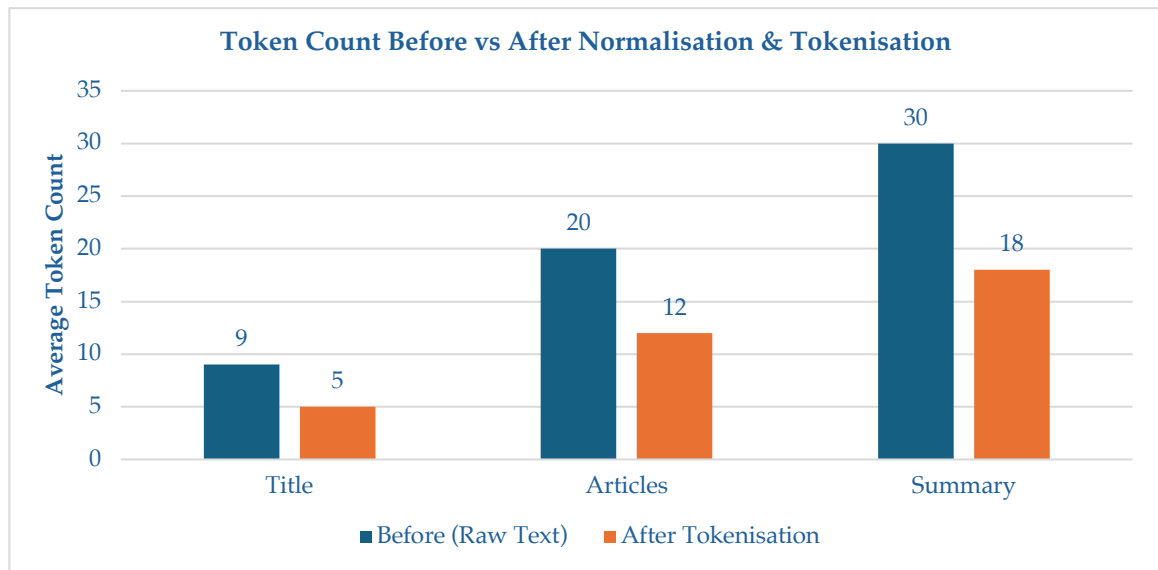


Chart 1.3: Token count before & after normalisation & tokenisation.

As further steps, below pictures demonstrates how lemmatisation and sentence segmentation is performed on the dataset considering only first 5 rows of the dataset.



Figure1.2: Lemmatisation clustering: word forms reduced to base lemmas.

As a final step of data preprocessing, embedding is generated on three columns of Title, Articles, and Summary using “sentence transformers”.

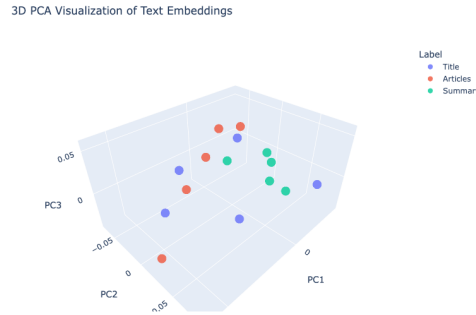


Figure1.3: 3D PCA visualisation of text embeddings.¹⁴

The 3D picture demonstrates top three principal components (PC) that shows the maximum variance and biggest differences between text fields in the mentioned three columns. Title points are grouped together, which means they are semantically consistent. Article points spread out, which means legal references vary across different cases. Summary points are close to article points that reflects summaries are providing description and application of articles on different cases. There are overlapping and completely distinct clusters, which shows similarity and semantic distinction in the information included in the mentioned three columns.

To demonstrate the dataset preprocessing more, an example query is pre-processed and shown below:

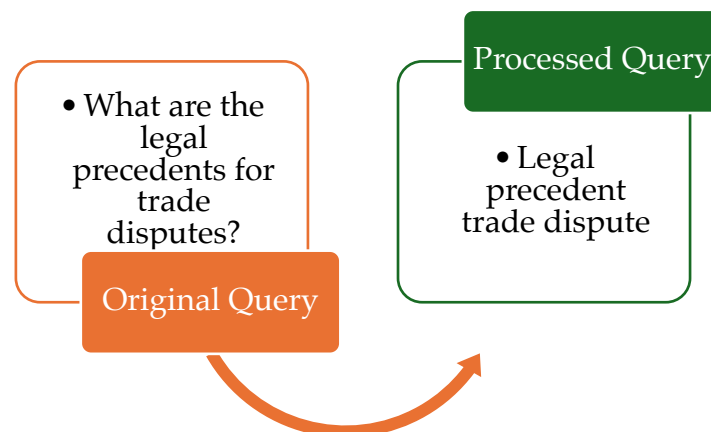


Figure 1.4: Pre-processed query example.

At the end, the pre-processed dataset with its embeddings is saved for implementation of models as a “csv” file using “numpy”.

¹⁴ Python Software Foundation. Python Language Reference, version 3. Available at: <https://www.python.org/>

2 Chapter Two

2.1 Baseline Information Retrieval System

The primary goal of developing a baseline IR system is to retrieve and rank a set of initial case laws from the WTO dataset using traditional IR techniques. The retrieved and ranked documents are based on the user's query. Two methods/models are used in this system: BM25 and Sentence-BERT. BM25 has ranked case laws based on term and inverse frequency in the case laws. However, Sentence-BERT has converted the query and case laws into vector embeddings and calculated similarity accordingly. Sentence-BERT also captured semantic meaning, which makes its ranking results more accurate and reliable than BM25 in terms of legal context.¹⁵

Features	BM25	Sentence-BERT
Retrieval	Keyword	Semantic meaning
Speed	Fast & efficient	Slow, more computational
Understanding Synonyms	No	Yes
Context Awareness	No	Yes, because captures meaning
Paraphrasing	Actual match only	Strong generalisation
Best Performance	Simple keyword queries	Complex legal or semantic queries

*Table 2.1: BM25 vs. Sentence-BERT comparison.*¹⁶

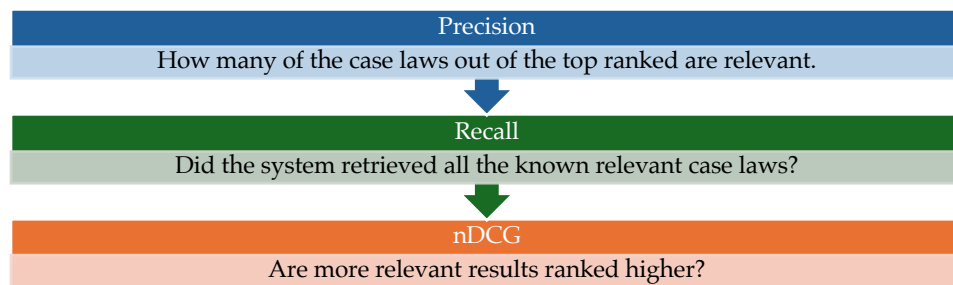
The dataset used in baseline IR system is the pre-processed dataset. Three important columns are used to perform BM25 and Sentence-BERT: Title-tokenised, Articles-lemmatised, and Summary-lemmatised. The Case IDs are also used to identify each ranking. For implementing the baseline IR system, three queries are tested. These queries are designed to consider three types of users: a lawyer, an academic, and a policy maker. Each user might be interested in different information from the case laws and therefore queries differ accordingly. The influence and personalisation demand of the user profile will be discussed in more depth in the next chapter. The queries are:

1. US – China case laws using article VIII of GATT (Lawyer).
2. Case laws involving European Communities as complainant and United States as respondent (Academic).
3. Case laws involving Australia with ASCM article application (Policy Maker).

¹⁵ Eds, H. *et al.* (2022) 'Proceedings of the 3rd Workshop on Patent Text Mining and Semantic Technologies', Patent Sem Tech, 15 July 2022, Madrid.

¹⁶ Askari, A. Abolghasemi, A. *et al.* (2023) 'Injecting the BM25 Score as Text Improves BERT-Based Re-rankers', ECIR, 17 March. https://doi.org/10.1007/978-3-031-28244-7_5

After testing these queries, an evaluation is performed to check the accuracy of the results for each method/model (BM25 and Sentence-BERT). The evaluation is based on Precision, Recall, and nDCG.¹⁷ Libraries and functions used are “sklearn.metrics”, “ndcg_score”, and “numpy”. To do this evaluation and check the performance of both models, the ground truth case IDs are defined based on expert knowledge and manually. Then, the performance of both models is compared to expert knowledge-based findings.



2.1.1 BM25:

To perform BM25, “rank_bm25” package is used. Libraries and functions used are: “pandas”, “nltk”, “word_tokenize” and “BM25Okapi”. First, the input of case laws and the query are tokenized. Second, the scores for ranking case laws are calculated. And last, top 5 retrieved case laws are displayed only with Case IDs and Titles. Meanwhile, an evaluation of BM25 is performed compared to the ground truth. It is worth mentioning that the order of ground truth case laws does not determine any ranking among them. The results for all three queries are as below:

¹⁷ Deng, L. *et al.* (2013) ‘Deep stacking networks for information retrieval’, *IEEE International Conference on Acoustics, Speech and Signal Processing*, Vancouver, pp. 3153-3157.

Query 1: US – China Case Laws using Article VIII of GATT			Query 2: Case Laws involving European Communities as Complainant and United States as Respondent			Query 3: Case Laws involving Australia with ASCM Article Application		
1	Ds395	China – Raw Materials	1	Ds266	EC – Export Subsidies on Sugar	1	Ds126A	Australia - Automotive Leather II (Article 21.5 – US)
2	Ds422	US – Shrimp and Sawblades (China)	2	Ds165	US – Certain EC Products	2	Ds126	Australia – Automotive Leather II
3	Ds432	China – Rare Earths	3	Ds34	Turkey - Textiles	3	Ds18A	Australia – Salmon (Article 21.5 – Canada)
4	Ds339	China – Auto Parts	4	Ds62	EC – Computer Equipment	4	Ds353	US – Large Civil Aircraft (2 nd Complaint)
5	Ds437	US – Countervailing Measures (China)	5	Ds108B	US – FSC (Article 21.5 – EC II)	5	Ds367	Australia – Apples

Table 2.2: Queries rankings in BM25.

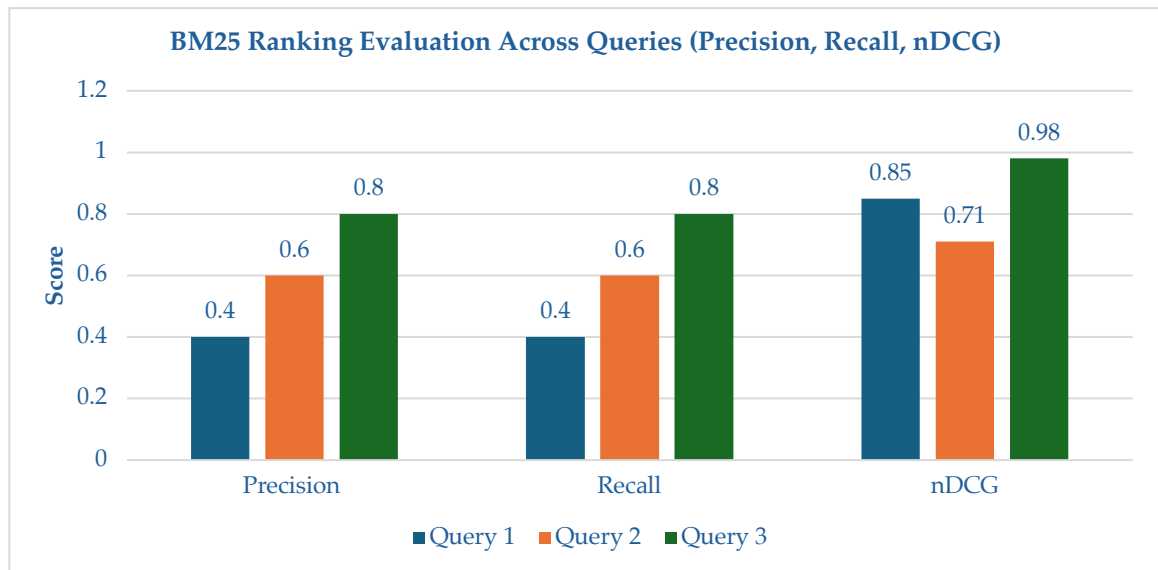


Chart 2.1: BM25 ranking evaluation.

Query 1: 2 out of top 5 documents are relevant (Precision), which makes 40% of all relevant case laws (Recall). The ranking quality is strong, meaning most relevant case laws are ranked near the top (nDCG).

Query 2: 3 out of top 5 case laws are relevant (Precision), which makes 60% of all relevant case laws (Recall). The ranking quality is moderate, meaning more than half of the relevant case laws are ranked near the top (nDCG).

Query 3: 4 out of top 5 case laws are relevant (Precision), which makes 80% of all relevant case laws (Recall). The ranking quality is ideal, meaning nearly all the relevant case laws are ranked near the top (nDCG).

2.1.2 Sentence BERT:

Similar to BM25, Sentence-BERT is applied on the same three queries. The “sentence-transformers” and “torch” packages are used along with “SentenceTransformer, util” library. First, an embedding of the case laws and the query is created. Second, the cosine similarity is computed, and top 5 indices are selected. Lastly, top 5 Case IDs and Titles are extracted based on semantic meaning. An evaluation of Sentence-BERT performance with Precision, Recall, and nDCG factors carried out to check the quality of Sentence-BERT result against ground truth. The results are demonstrated below:

Query 1: US – China Case Laws using Article VIII of GATT			Query 2: Case Laws involving European Communities as Complainant and United States as Respondent			Query 3: Case Laws involving Australia with ASCM Article Application		
1	Ds395	China – Raw Materials	1	Ds62	EC – Computer Equipment	1	Ds126	Australia – Automotive Leather II
2	Ds432	China – Rare Earths	2	Ds108B	US – FSC (Article 21.5 – EC II)	2	Ds126A	Australia - Automotive Leather II (Article 21.5 – US)
3	Ds27B	EC – Bananas III (Article 21.5 – US)	3	Ds231	EC - Sardines	3	Ds367	Australia – Apples
4	Ds517	China - TRQS	4	Ds577	US – Ripe Olives from Spain	4	Ds18A	Australia – Salmon (Article 21.5 – Canada)
5	Ds510	US – Renewable Energy (India)	5	Ds34	Turkey – Textiles	5	Ds529	Australia – Anti- Dumping Measures on Paper

Table 2.3: Queries rankings in Sentence-BERT.

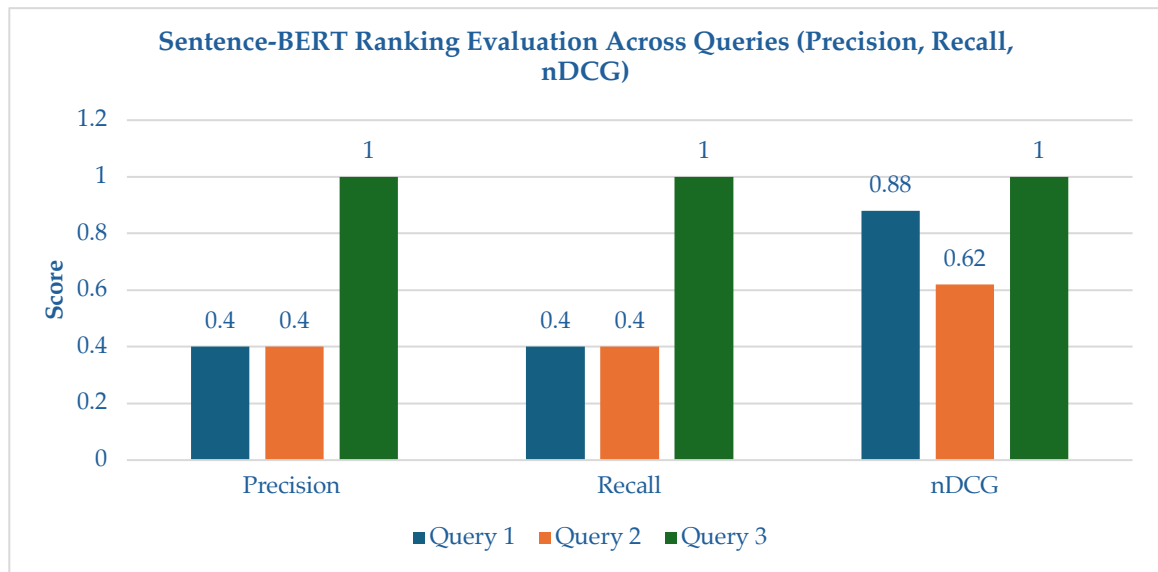


Chart 2.2: Sentence-BERT ranking evaluation.

Query 1: 2 out of top 5 documents are relevant (Precision), which makes 40% of all relevant case laws (Recall). The ranking quality is strong, meaning most relevant case laws are ranked near the top (nDCG).

Query 2: 2 out of top 5 case laws are relevant (Precision), which makes 40% of all relevant case laws (Recall). The ranking quality is low, meaning less than half of the relevant case laws are ranked near the top (nDCG).

Query 3: 5 out of top 5 case laws are relevant (Precision), which makes 100% of all relevant case laws (Recall). The ranking quality is perfect, meaning all the relevant case laws are ranked near the top (nDCG).

As seen in the tables and charts above, the retrieved and ranked case laws differ in BM25 and Sentence-BERT. The evaluation results slightly differ in both systems. Both has shown relatively reliable results; however, Sentence-BERT seems to be more accurate due to focus on semantic meaning and its designation for complex legal queries.¹⁸ Therefore, to add more personalised features such as justifying why a specific case law is relevant to the query in terms of context and ranking and to predict an outcome for a similar case considering different aspects of a user profile, an LLM is constituted in the next chapter.

¹⁸ Zhang, Z. et al. (2020) 'Semantics-Aware BERT for Language Understanding', *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05), pp. 9628-9635. <https://doi.org/10.1609/aaai.v34i05.6510>

3 Chapter Three

3.1 Large Language Model

LLM is used to improve the outcome of the Baseline IR System and make it more personalised. LLM ranks the top 5 case laws according to query and user profile. It provides a short natural language justification of why each ranking is relevant and makes a prediction to explain what the potential result would be if the user has a similar case law to deal with. As mentioned in previous chapter, queries differ based on different information each user is interested in. This means beside query the user profile information given to LLM has a great impact on the personalised outcome that LLM provides. LLM analyses the query based on the interests of the user in its ranking, justification, and prediction performance. The user profile information used for LLM in this project are:



Figure 3.1: Example of User Profile information.

LLM is performed using OpenAI API's GPT-4 model, which uses an individual API key. The packages used to perform LLM are: "openai", "pandas", "numpy", and "sklearn". The libraries used are: "json", "time" and "math". Similar to baseline IR system, the pre-processed dataset is used for LLM as well focusing on three main

columns such as Title, Articles-lemmatised, and Summary-lemmatised including Case IDs to identify the rankings. LLM is performed in four below steps:

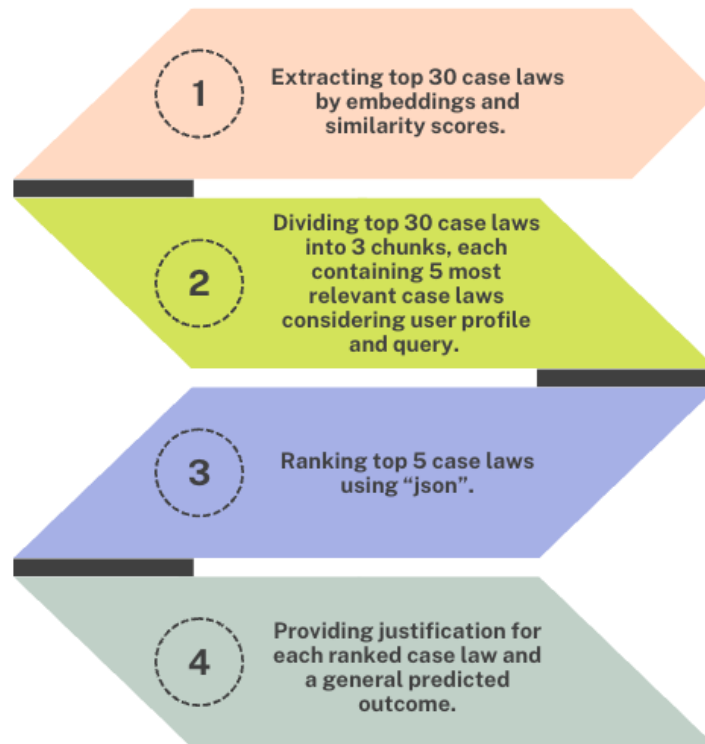


Figure 3.2: Steps of LLM.

In the first step, an embedding is created for all WTO dataset, query, and user profile, and similarity scores are calculated accordingly to extract top 30 case laws among 111. The reason for shortening the WTO dataset is that the GPT-4 of OpenAI API has a limit for input tokens per minute. Inputting all 111 case laws at once goes over the limit and does not perform. In the second step, beside chunking function, a prompt is built for user profile and query that specifies the task for GPT-4 with a message about its role and determines the temperature (0.0) to be used in generating a proper “json” response. In the third step, using the 15 case laws from chunks, the prompt built in step 2 with different task, the same query and user profile, and the same characteristics for GPT-4, a final top 5 case laws are ranked. In step four, GPT-4 preserves the original ranking of top 5 case laws while generating justifications and prediction using the same query, user profile, prompt (but different task), and GPT-4 characteristics. It is worth mentioning that GPT-4 naturally alters the wording of the justifications and prediction every time the code runs without any alteration to ranking of case laws or context of justifications and prediction. Also, the tasks identified in each step has a vital

role in every response generated. All three queries are visualised from step 2 - 4 with two natural alterations of GPT-4 below. The results for step 1 will be in Appendix.

Query 1: US – China Case Laws using Article VIII of GATT

User Profile Information for Query 1:

- **Name:** Madiha Ahmadi
- **Role:** International Trade Lawyer
- **Interests:** GATT Article VIII, US – China Disputes
- **Jurisdiction Focus:** United States, China
- **Explanation Level:** International Trade Law

Step 2: Chunking								
Chunk 1			Chunk 2			Chunk 3		
1	Ds437	US – Countervailing Measures (China)	1	Ds492	US – Poultry Meat (China)	1	Ds384	US – Cool (Article 21.5 – Canada and Mexico)
2	Ds471	US – Anti-Dumping Methodologies (China)	2	Ds152	US – Section 301 Trade Act	2	Ds108A	US – FSC (Article 21.5 – EC)
3	Ds437A	US – Countervailing Measures (China) (Article 21.5 – China)	3	Ds432	China – Rare Earths	3	Ds294	US – Zeroing (EC)
4	Ds395	China – Raw Materials	4	Ds339	China – Auto Parts	4	Ds194	US – Export Restraints
5	Ds440	China – Autos (US)	5	Ds316B	(Article 21.5 – US)	5	Ds464	US – Washing Machines

Table 3.1: LLM chunking result for query 1.

Step 2 – Prompt

Message: You are a legal assistant helping an international trade lawyer.

Task: From the list below, select the 5 most relevant cases for each chunk based on legal reasoning and relevance to the query.

Step 3 & 4 – Alteration 1: Top 5 Case Laws, Justifications & Prediction			
General Predicted Outcome: Based on these precedents, a WTO panel might rule that the US and China must ensure their countervailing measures and anti-dumping methodologies are consistent with their obligations under the WTO Agreement and the DSU. The panel might also rule that the US and China must provide sufficient evidence to support their determinations and must base their injury determinations on an objective examination based on positive evidence.			
1	Ds437	US – Countervailing Measures (China)	This case is relevant as it deals with the US Department of Commerce's (USDOC) determination of Chinese state-owned enterprises as public bodies, which is a key aspect of GATT Article VIII. The case also discusses the use of in-country private prices in China as benchmarks for benefit analysis, which could be relevant in a US-China dispute involving Article VIII.
2	Ds471	US – Anti-Dumping Methodologies (China)	This case is relevant as it discusses the US's anti-dumping methodologies in relation to China, which could be relevant in a US-China dispute involving Article VIII. The case also discusses the use of zeroing in dumping margin calculations, which could be relevant in a US-China dispute involving Article VII I.
3	Ds437A	US – Countervailing Measures (China) (Article 21.5 – China)	This case is relevant as it discusses the USDOC's determination of Chinese state-owned enterprises as public bodies, which is a key aspect of GATT Article VIII. The case also discusses the use of out-of-country prices as benchmarks for benefit analysis, which could be relevant in a US-China dispute involving Article VIII.
4	Ds152	US – Section 301 Trade Act	This case is relevant as it discusses the US's unilateral determination of measures inconsistent with WTO obligations, which could be relevant in a US-China dispute involving Article VIII. The case also discusses the US's obligations under the WTO Agreement and the DSU, which could be relevant in a US-China dispute involving Article VIII.
5	Ds440	China – Autos (US)	This case is relevant as it discusses China's anti-dumping and countervailing duty measures against the US, which could be relevant in a US-China dispute involving Article VIII. The case also discusses China's injury determination, which could be relevant in a US-China dispute involving Article VIII.

Table 3.2: Top 5 case laws, LLM generated justifications, and prediction for query 1 – alteration 1.

Step 3 – Prompt

Message: You are a legal assistant helping an international trade lawyer.

Task: From the list of 15 case laws below, select the 5 most relevant to this query. Provide ranked list from most to least relevant based on legal context, query and user interests.

Step 3 & 4 – Alteration 2: Top 5 Case Laws, Justifications & Prediction			
General Predicted Outcome: Based on these precedents, a WTO panel might rule that both the US and China have obligations to adhere to WTO rules in their trade practices. This includes the use of appropriate methodologies in anti-dumping investigations, the proper determination of public bodies, and the use of in-country prices as benchmarks for benefit analysis. The panel might also rule against unilateral actions by member states and require the disclosure of essential facts in anti-dumping and countervailing duty investigations.			
1	Ds437	US – Countervailing Measures (China)	This case is relevant as it deals with the determination of public bodies and the use of in-country prices as benchmarks for benefit analysis. It provides insight into how the WTO views the role of state-owned enterprises and the use of domestic prices in determining benefits.
2	Ds471	US – Anti-Dumping Methodologies (China)	This case is significant as it addresses the methodologies used by the US in anti-dumping investigations involving China. It provides a precedent on how the WTO views the use of different comparison methodologies and the requirement to consider evidence of non-target prices.
3	Ds437A	US – Countervailing Measures (China) (Article 21.5 – China)	This case is an appeal of the ds437 case and further clarifies the definition of public bodies and the use of in-country prices as benchmarks for benefit analysis. It provides additional insight into how the WTO views these issues.
4	Ds152	US – Section 301 Trade Act	This case is relevant as it deals with the US's unilateral determination of measures inconsistent with WTO obligations. It provides a precedent on how the WTO views unilateral actions by member states.
5	Ds440	China – Autos (US)	This case is significant as it addresses China's anti-dumping and countervailing duty measures on certain automobiles from the US. It provides a precedent on how the WTO views the use of confidential information and the disclosure of essential facts in anti-dumping and countervailing duty investigations.

Table 3.3: Top 5 case laws, LLM generated justifications, and prediction for query 1 – alteration 2.

Step 4 – Prompt

Message: You are a legal assistant helping an international trade lawyer.

Task: Below are 5 WTO case law summaries retrieved for this query. For each of the 5 selected cases, provide a professional legal justification for its relevance, written for a {user_profile['explanation_level']} explanation level. Predict what kind of ruling a WTO panel might make in a similar dispute based on these precedents.

Query 2: Case Laws Involving European Communities (EC) as Complainant and United States as Respondent

User Profile Information for Query 2:

- **Name:** Madiha Ahmadi
- **Role:** Academic
- **Interests:** EC – US disputes, Complainant, Respondent
- **Jurisdiction Focus:** United States, European Communities
- **Explanation Level:** International Trade Interests

Step 2: Chunking								
Chunk 1			Chunk 2			Chunk 3		
1	Ds353A	US – Large Civil Aircraft (2 nd Complainant) (Article 21.5 – EU)	1	Ds353	US – Large Civil Aircraft (2 nd Complainant)	1	Ds69	EC – Poultry
2	Ds165	US – Certain EC Products	2	Ds108B	US – FSC (Article 21.5 – EC II)	2	Ds194	US – Export Restraints
3	Ds108A	US – FSC (Article 21.5 – EC)	3	Ds27B	EC – Bananas III (Article 21.5 – US)	3	Ds258	US – Steel Safeguard
4	Ds577	US – Ripe Olives from Spain	4	Ds294	US – Zeroing (EC)	4	Ds277	US – Softwood Lumber VI
5	Ds62	EC – Computer Equipment	5	Ds316B	(Article 21.5 – US)	5	Ds234	US – Offset Act (Byrd Amendments)

Table 3.4: LLM chunking result for query 2.

Step 2 – Prompt

Message: You are a Ph.D. student helping a law academic professor.

Task: From the list below, select the 5 most relevant cases for each chunk based on legal reasoning and relevance to the query.

Step 3 & 4 – Alteration 1: Top 5 Case Laws, Justifications & Prediction			
General Predicted Outcome: Based on these precedents, a WTO panel might rule in favour of the complainant (EC) if it can demonstrate that the respondent (US) has implemented measures that are inconsistent with its obligations under the relevant WTO agreements. However, the panel's ruling would likely depend on the specific facts and circumstances of the dispute, as well as the interpretation and application of the relevant WTO provisions.			
1	Ds353A	US – Large Civil Aircraft (2 nd Complainant) (Article 21.5 – EU)	This case is relevant as it involves a dispute over subsidies provided by the US to Boeing, a large civil aircraft manufacturer. The European Communities (EC) argued that these subsidies were inconsistent with the Agreement on Subsidies and Countervailing Measures (ASCM). The case provides insights into how the WTO interprets and applies the ASCM, particularly in relation to the definition of a subsidy, the assessment of benefits conferred, and the determination of adverse effects.
2	Ds108A	US – FSC (Article 21.5 – EC)	This case involves a dispute over the US's Foreign Sales Corporation (FSC) scheme, which the EC argued constituted a prohibited export subsidy under the ASCM. The case is relevant as it provides insights into how the WTO interprets and applies the ASCM, particularly in relation to the definition of a financial contribution, the determination of export contingency, and the application of the double taxation exception.
3	Ds108B	US – FSC (Article 21.5 – EC II)	This case is a follow-up to ds108A and involves a dispute over the US's continued implementation of the FSC scheme despite previous WTO rulings. The case is relevant as it provides insights into how the WTO handles cases where a member fails to fully implement the DSB's recommendations and rulings.
4	Ds294	US – Zeroing (EC)	This case involves a dispute over the US's use of "zeroing" methodology in its anti-dumping investigations. The EC argued that this methodology was inconsistent with the Anti-Dumping Agreement (ADA) and the General Agreement on Tariffs and Trade (GATT). The case is relevant as it provides insights into how the WTO interprets and applies the ADA and GATT, particularly in relation to the calculation of dumping margins and the imposition of anti-dumping duties.
5	Ds165	US – Certain EC Products	This case involves a dispute over the US's imposition of additional bond requirements on certain EC products. The EC argued that these requirements were inconsistent with the GATT and the Dispute Settlement Understanding (DSU). The case is relevant as it provides insights into how the WTO interprets and applies the GATT and DSU, particularly in relation to the most-favoured-nation treatment, the schedule of concessions, and the prohibition of unilateral determinations.

Table 3.5: Top 5 case laws, LLM generated justifications, and prediction for query 2 – alteration 1.

Step 3 – Prompt

Message: You are a Ph.D. student helping a law academic professor.

Task: From the list of 15 case laws below, select the 5 most relevant to this query. Provide ranked list from most to least relevant based on legal context, query and user interests.

Step 3 & 4 – Alteration 2: Top 5 Case Laws, Justifications & Prediction

General Predicted Outcome: Based on these precedents, a WTO panel might rule in a similar dispute that the measure in question is inconsistent with the relevant WTO agreement if it can be shown that the measure confers a benefit that is contingent on export performance or causes adverse effects to the interests of other members. The panel might also rule that the member has failed to fully implement the DSB's recommendations and rulings if it continues to maintain the measure despite a previous ruling that it is inconsistent with the relevant WTO agreement. However, the panel's ruling would depend on the specific facts and circumstances of the case.

1	Ds353A	US – Large Civil Aircraft (2 nd Complainant) (Article 21.5 – EU)	This case is relevant as it involves a dispute over subsidies provided by the US to Boeing, a large civil aircraft manufacturer. The European Communities (EC) argued that these subsidies were inconsistent with the Agreement on Subsidies and Countervailing Measures (ASCM). The case provides insights into how the WTO interprets and applies the ASCM, particularly in relation to the definition of a subsidy, the assessment of benefits conferred, and the determination of adverse effects.
2	Ds108A	US – FSC (Article 21.5 – EC)	This case involves a dispute over the US's Foreign Sales Corporation (FSC) scheme, which the EC argued constituted a prohibited export subsidy under the ASCM. The case is relevant as it provides insights into how the WTO interprets and applies the ASCM, particularly in relation to the definition of an export subsidy and the exceptions for measures to avoid double taxation.
3	Ds108B	US – FSC (Article 21.5 – EC II)	This case is a follow-up to ds108A and involves a dispute over the US's continued implementation of the FSC scheme despite the WTO's previous ruling that it constituted a prohibited export subsidy. The case is relevant as it provides insights into how the WTO handles cases where a member fails to fully implement the DSB's recommendations and rulings.
4	Ds294	US – Zeroing (EC)	This case involves a dispute over the US's use of "zeroing" in its anti-dumping calculations, which the EC argued was inconsistent with the Anti-Dumping Agreement (ADA) and the General Agreement on Tariffs and Trade (GATT). The case is relevant as it provides insights into how the WTO interprets and applies the ADA and GATT, particularly in relation to the calculation of dumping margins.
5	Ds165	US – Certain EC Products	This case involves a dispute over the US's imposition of additional bond requirements on certain EC products, which the EC argued was inconsistent with the GATT and the Dispute Settlement Understanding (DSU). The case is relevant as it provides

			ides insights into how the WTO interprets and applies the GATT and DSU, particularly in relation to the most-favoured-nation treatment and the prohibition on unilateral determinations.
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Table 3.6: Top 5 case laws, LLM generated justifications, and prediction for query 2 – alteration 2.

Step 4 – Prompt

Message: You are a Ph.D. student helping a law academic professor.

Task: Below are 5 WTO case law summaries retrieved for this query. For each of the 5 selected cases, provide a professional legal justification for its relevance, written for a {user_profile['explanation_level']} explanation level. Predict what kind of ruling a WTO panel might make in a similar dispute based on these precedents.

Query 3: Case Laws Involving Australia with ASCM Article Application

User Profile Information for Query 3:

- **Name:** Madiha Ahmadi
- **Role:** Policy Maker
- **Interests:** ASCM Articles, Legal Reasoning, Compliance to WTO Rulings
- **Jurisdiction Focus:** Australia
- **Explanation Level:** Policies.

Step 2: Chunking									
Chunk 1			Chunk 2			Chunk 3			
1	Ds18A	Australia – Salmon (Article 21.5 – Canada)	There are no relevant case laws in this chunk.			1	Ds436A	US – Carbon Steel (India)	
2	Ds126	Australia – Automotive Leather II				2	Ds62	EC – Computer Equipment	
3	Ds126A	Australia – Automotive Leather II (Article 21.5 – US)				3	Ds135	EC - Asbestos	
4	Ds367	Australia – Apples				4	Ds456	India – Solar Cells	
5	Ds529	Australia – Anti-Dumping Measures on Paper				5	Ds165	US – Certain EC Products	

Table 3.7: LLM chunking result for query 3.

Step 2 – Prompt

Message: You are a policy assistant helping a senior policy analyst.

Task: From the list below, select the 5 most relevant cases for each chunk based on legal reasoning and relevance to the query.

Step 3 & 4 – Alteration 1: Top 5 Case Laws, Justifications & Prediction			
General Predicted Outcome: Based on these precedents, a WTO panel is likely to take a strict approach to interpreting the ASCM and other related agreements. The panel is likely to find in favour of the complainant if it can demonstrate that the respondent has provided subsidies contingent upon export performance or has otherwise engaged in unfair trade practice. The panel is also likely to require full compliance with its rulings, including the withdrawal of prohibited subsidies.			
1	Ds126	Australia – Automotive Leather II	This case is relevant as it involves the application of ASCM Article 3.1(a) which prohibits subsidies contingent upon export performance. The panel found that the payments made under the grant contract were subsidies prohibited under Article 3.1(a) as they were tied to export performance. This case provides a clear interpretation of what constitutes an export subsidy under ASCM.
2	Ds126A	Automotive Leather II (Article 21.5 – US)	This case is a follow-up to the previous one and provides further clarification on the obligations of a member state under ASCM Article 4.7 to withdraw prohibited subsidies. The panel found that repayment in full of the prohibited subsidy was necessary in this case, providing guidance on the measures required to comply with a ruling under ASCM.
3	Ds18A	Australia – Salmon (Article 21.5 – Canada)	Although this case primarily involves the SPS Agreement, it is relevant as it provides insight into the WTO panel's approach to assessing compliance with its rulings. The panel found that Australia had not complied with its previous ruling, demonstrating the importance of full compliance with WTO rulings.
4	Ds367	US – Apples	This case is relevant as it involves the application of the SPS Agreement, which is closely related to the ASCM in terms of its focus on fair trade practices. The panel's findings provide insight into the interpretation of the SPS Agreement, which could be useful in understanding the ASCM.
5	Ds529	Australia – Anti-Dumping Measures on Paper	This case is relevant as it involves the application of the Anti-Dumping Agreement, which is closely related to the ASCM in terms of its focus on preventing unfair trade practices. The panel's findings provide insight into the interpretation of the Anti-Dumping Agreement, which could be useful in understanding the ASCM.

Table 3.8: Top 5 case laws, LLM generated justifications, and prediction for query 3 – alteration 1.

Step 3 – Prompt

Message: You are a policy assistant helping a senior policy analyst.

Task: From the list of 15 case laws below, select the 5 most relevant to this query. Provide ranked list from most to least relevant based on legal context, query and user interests.

Step 3 & 4 – Alteration 2: Top 5 Case Laws, Justifications & Prediction			
General Predicted Outcome: Based on these precedents, a WTO panel in a similar dispute might rule that subsidies contingent upon export performance are prohibited under ASCM Article 3.1(a) and that the withdrawal of such subsidies under ASCM Article 4.7 encompasses repayment. The panel might also rule that risk assessments under the SPS Agreement must meet certain requirements and that discrimination is prohibited. Finally, the panel might rule that dumping determinations and cost calculations under the Anti-Dumping Agreement must meet certain requirements.			
1	Ds126	Australia – Automative Leather II	This case is relevant as it involves the application of ASCM Article 3.1(a) which prohibits subsidies contingent upon export performance. The panel's interpretation of the term "subsidy" and its application to the facts of the case provides valuable insight into how such provisions might be applied in future disputes.
2	Ds126A	Automative Leather II (Article 21.5 – US)	This case is a follow-up to the previous one and provides further clarification on the application of ASCM Article 4.7, which requires the withdrawal of prohibited subsidies. The panel's interpretation of "withdrawal" as encompassing repayment of the subsidy is a significant legal interpretation that could impact future cases.
3	Ds18A	Australia – Salmon (Article 21.5 – Canada)	This case is relevant as it involves the application of the SPS Agreement, which is closely related to the ASCM in terms of its focus on trade restrictions. The panel's interpretation of the requirements for a risk assessment under Article 5.1 of the SPS Agreement could have implications for the interpretation of similar provisions in the ASCM.
4	Ds367	US – Apples	This case is relevant as it involves the application of the SPS Agreement and provides insight into how the panel interprets and applies the requirements for risk assessment and the prohibition on discrimination. These interpretations could have implications for similar provisions in the ASCM.
5	Ds529	Australia – Anti-Dumping Measures on Paper	This case is relevant as it involves the application of the Anti-Dumping Agreement, which is closely related to the ASCM in terms of its focus on trade restrictions. The panel's interpretation of the requirements for dumping determinations and cost calculations could have implications for similar provisions in the ASCM.

Table 3.9: Top 5 case laws, LLM generated justifications, and prediction for query 3 – alteration 2.

Step 4 – Prompt

Message: You are a policy assistant helping a senior policy analyst.

Task: Below are 5 WTO case law summaries retrieved for this query. For each of the 5 selected cases, provide a professional legal justification for its relevance, written for a {user_profile['explanation_level']} explanation level. Predict what kind of ruling a WTO panel might make in a similar dispute based on these precedents.

An evaluation of the LLM results is also performed against expert knowledge ground truth using “numpy” package. The factors are Precision, Recall, and nDCG.

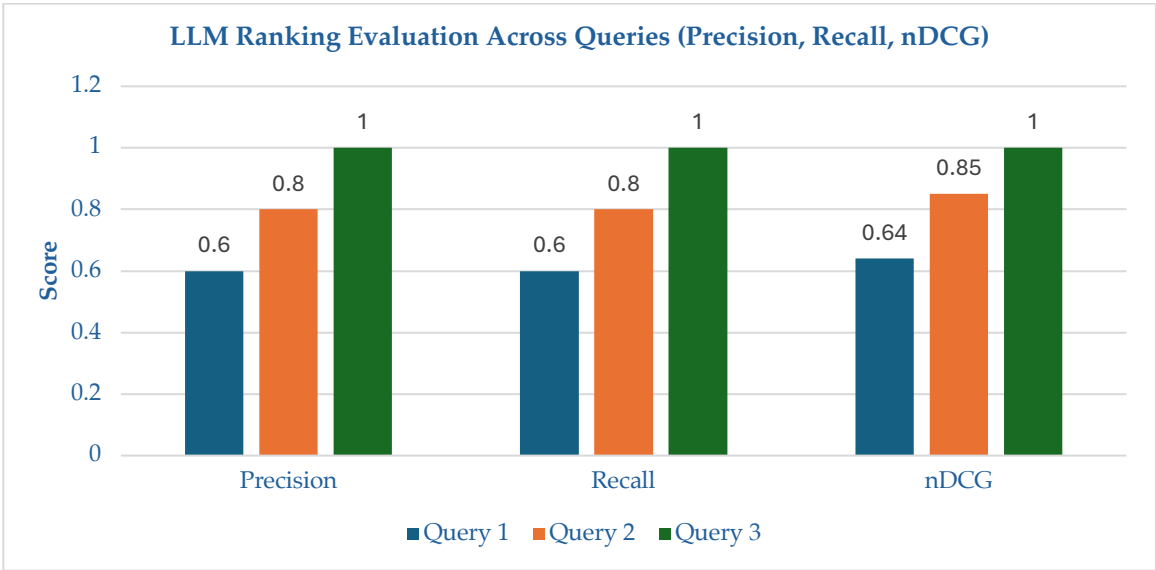


Chart 3.1: LLM ranking evaluation.

Query 1: 3 out of top 5 case laws are relevant (Precision), which makes 60% of all relevant case laws (Recall). The ranking quality is moderate, meaning more than half of the relevant case laws are ranked near the top (nDCG).

Query 2: 4 out of top 5 case laws are relevant (Precision), which makes 80% of all relevant case laws (Recall). The ranking quality is ideal, meaning nearly all the relevant case laws are ranked near the top (nDCG).

Query 3: 5 out of top 5 case laws are relevant (Precision), which makes 100% of all relevant case laws (Recall). The ranking quality is perfect, meaning all the relevant case laws are ranked near the top (nDCG).

4 Further Discussion

There are further steps that can be taken to make this project more practical, beneficial, and usable for law firms and/or individual legal practitioners. Also, it makes it user friendly for those who do not technically do coding. These steps need more resources and time, which were out of capacity of the current thesis project.

4.1 Legal Case Assistant Chatbot

A deployed chatbot can be produced via different platforms such as Llamaindex, LangChain, Gradio or Streamlit, which are LLM-Based Tools. The chatbot would provide the personalised rankings, justifications, and prediction like LLM. The chatbot characteristics can be the followings:

1. An interface that can have the space to enter query and user profile information.
2. The space to give a message to GPT-4 about its role and task.
3. Allow follow-up questions after the first response.
4. Have citation linking to link the provided response to other case law citations in order to improve credibility.
5. Option to save favourite cases and responses.
6. An evaluation and feedback loop to allow users give comments about its performance in providing credible answers.
7. Works well with more complex legal terminology and queries.
8. Works well with bigger databases.

Legal Case Assistant Chatbot

Query

Enter your query

User Profile

Name

Role

Interests

GPT-4 is a legal case assistant. Ask it a question about case law.



China – Rare Earths (DS432) ⁵ is directly relevant to the query, as it concerns a dispute between the US and China regarding export duties on rare earths, and the applicability of [GATT Article Vill](#).

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Figure 4.1: ChatGPT-generate image visualising the interface of a chatbot.

5 Conclusion

Legal professionals such as lawyers, scholars, policy makers, etc face a significant challenge due to vast amount and complexity of legal documents. This includes retrieving relevant case laws or rulings, lack of personalised search tools, and the absence of explainability in search results. This project addresses these challenges by developing a personalised IR system that provides an initial document retrieval, and it leverages LLMs to refine the ranking of retrieved documents plus generating a natural language explainability feature with prediction. This system improves retrieval accuracy by tailoring results based on user preferences. It bridges the gap between traditional keyword-based legal search and modern AI-driven retrieval, making legal research more efficient, precise, and user-centric. It also improves decision making in the legal field. Similar to what ease and support AI brought to other professions, AI can support legal practice to make information retrieval easy. This could be one of the AI improvements to legal field.

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Appendices

All the code are available in the following link:

https://github.com/mahamdi170/MA_Law_AI.git

Appendix A: LLM Top 30 Cases Retrieved via Embeddings for Query 1

Case ID	Title
ds437	US – COUNTERVAILING MEASURES (CHINA) ¹
ds471	US – ANTI-DUMPING METHODOLOGIES (CHINA) ¹
ds437A	US – COUNTERVAILING MEASURES (CHINA) (ARTICLE 21.5 – CHINA) ¹
ds395	CHINA – RAW MATERIALS ¹
ds165	US – CERTAIN EC PRODUCTS ¹
ds322A	US – ZEROING (JAPAN) (ARTICLE 21.5 – JAPAN) ¹
ds440	CHINA – AUTOS (US) ¹
ds460	CHINA – HP-SSST (JAPAN/EUROPEAN UNION) ¹
ds258	US – STEEL SAFEGUARD
ds285A	US – GAMBLING (ARTICLE 21.5 – ANTIGUA AND BARBUDA) ¹
ds492	EU – POULTRY MEAT (CHINA) ¹
ds344	US – STAINLESS STEEL (MEXICO) ¹
ds152	US – SECTION 301 TRADE ACT ¹
ds322	US – ZEROING (JAPAN) ¹
ds510	US – RENEWABLE ENERGY (INDIA) ¹
ds432	CHINA – RARE EARTHS ¹
ds381A	US – TUNA II (ARTICLE 21.5 – MEXICO) ¹
ds339	CHINA – AUTO PARTS ¹
ds257A	US – SOFTWOOD LUMBER IV (ARTICLE 21.5 – CANADA) ¹
ds316B	(ARTICLE 21.5 – US) ¹
ds384	US – COOL (ARTICLE 21.5 – CANADA AND MEXICO) ¹
ds461A	COLOMBIA – TEXTILES ¹
ds44	JAPAN – FILM ¹
ds108A	US – FSC (ARTICLE 21.5 – EC) ¹

ds294	US – ZEROING (EC) ¹
ds194	US – EXPORT RESTRAINTS ¹
ds62	EC – COMPUTER EQUIPMENT ¹
ds464	US – WASHING MACHINES ¹
ds461	COLOMBIA – TEXTILES (ARTICLE 21.5 – PANAMA) ¹
ds69	EC – POULTRY ¹

Appendix B: LLM Top 30 Cases Retrieved via Embeddings for Query 2

Case ID	Title
ds353A	US – LARGE CIVIL AIRCRAFT (2ND COMPLAINT) (ARTICLE 21.5 – EU) ¹
ds165	US – CERTAIN EC PRODUCTS ¹
ds62	EC – COMPUTER EQUIPMENT ¹
ds437	US – COUNTERVAILING MEASURES (CHINA) ¹
ds285A	US – GAMBLING (ARTICLE 21.5 – ANTIGUA AND BARBUDA) ¹
ds322A	US – ZEROING (JAPAN) (ARTICLE 21.5 – JAPAN) ¹
ds577	US – RIPE OLIVES FROM SPAIN ¹
ds437A	US – COUNTERVAILING MEASURES (CHINA) (ARTICLE 21.5 – CHINA) ¹
ds257A	US – SOFTWOOD LUMBER IV (ARTICLE 21.5 – CANADA) ¹
ds108A	US – FSC (ARTICLE 21.5 – EC) ¹
ds353	US – LARGE CIVIL AIRCRAFT (2ND COMPLAINT) ¹
ds108B	US – FSC (ARTICLE 21.5 – EC II) ¹
ds316B	(ARTICLE 21.5 – US) ¹
ds231	EC – SARDINES ¹
ds27B	EC – BANANAS III (ARTICLE 21.5 – US) ¹
ds277A	US – SOFTWOOD LUMBER VI (ARTICLE 21.5 – CANADA) ¹
ds510	US – RENEWABLE ENERGY (INDIA) ¹
ds322	US – ZEROING (JAPAN) ¹
ds294	US – ZEROING (EC) ¹
ds471	US – ANTI-DUMPING METHODOLOGIES (CHINA) ¹

ds69	EC – POULTRY ¹
ds194	US – EXPORT RESTRAINTS ¹
ds258	US – STEEL SAFEGUARD
ds277	US – SOFTWOOD LUMBER VI ¹
ds234	US – OFFSET ACT (BYRD AMENDMENT) ¹
ds320	CANADA – CONTINUED SUSPENSION ¹
ds264	US – SOFTWOOD LUMBER V ¹
ds464	US – WASHING MACHINES ¹
ds381A	US – TUNA II (ARTICLE 21.5 – MEXICO) ¹
ds316A	EC AND CERTAIN MEMBER STATES – LARGE CIVIL AIRCRAFT ¹

Appendix C: LLM Top 30 Cases Retrieved via Embeddings for Query 3

Case ID	Title
ds18A	AUSTRALIA – SALMON (ARTICLE 21.5 – CANADA) ¹
ds126	AUSTRALIA – AUTOMOTIVE LEATHER II ¹
ds126A	AUSTRALIA – AUTOMOTIVE LEATHER II (ARTICLE 21.5 – US) ¹
ds367	AUSTRALIA – APPLES ¹
ds18	AUSTRALIA – SALMON ¹
ds437A	US – COUNTERVAILING MEASURES (CHINA) (ARTICLE 21.5 – CHINA) ¹
ds437	US – COUNTERVAILING MEASURES (CHINA) ¹
ds257A	US – SOFTWOOD LUMBER IV (ARTICLE 21.5 – CANADA) ¹
ds316B	(ARTICLE 21.5 – US) ¹
ds529	AUSTRALIA – ANTI-DUMPING MEASURES ON PAPER ¹
ds322A	US – ZEROING (JAPAN) (ARTICLE 21.5 – JAPAN) ¹
ds277A	US – SOFTWOOD LUMBER VI (ARTICLE 21.5 – CANADA) ¹
ds234	US – OFFSET ACT (BYRD AMENDMENT) ¹
ds108B	US – FSC (ARTICLE 21.5 – EC II) ¹
ds285A	US – GAMBLING (ARTICLE 21.5 – ANTIGUA AND BARBUDA) ¹

ds353A	US – LARGE CIVIL AIRCRAFT (2ND COMPLAINT) (ARTICLE 21.5 – EU)1
ds108A	US – FSC (ARTICLE 21.5 – EC)1
ds436	US – CARBON STEEL (INDIA) (ARTICLE 21.5 – INDIA)1
ds316A	EC AND CERTAIN MEMBER STATES – LARGE CIVIL AIRCRAFT1
ds353	US – LARGE CIVIL AIRCRAFT (2ND COMPLAINT)1
ds436A	US – CARBON STEEL (INDIA)1
ds62	EC – COMPUTER EQUIPMENT1
ds135	EC – ASBESTOS1
ds456	INDIA – SOLAR CELLS 1
ds165	US – CERTAIN EC PRODUCTS1
ds267A	US – UPLAND COTTON (ARTICLE 21.5 – BRAZIL)1
ds221	US – SECTION 129(C)(1) URAA1
ds268A	(ARTICLE 21.5 – ARGENTINA)1
ds268	US – OIL COUNTRY TUBULAR GOODS SUNSET REVIEWS1
ds384	US – COOL (ARTICLE 21.5 – CANADA AND MEXICO)1